

Deep Learning

A coherent, practice-led progression from neural foundations to responsible deployment

COURSE CODE	AI-XXX-XXXL (departmental placeholder retained)
CREDIT HOURS	3 (2+1)
WEEKLY CONTACT	2 lecture + 3 lab hours per week
PRIMARY ENVIRONMENT	Google Colab with PyTorch
TEACHING PATTERN	16 lectures x 2 hours; 16 laboratories x 3 hours

This revision preserves the approved credit/contact structure while strengthening prerequisites, sequencing, assessment evidence, book references, reproducibility, responsible AI, and deployment practice.

1. Course identity

Course title	Deep Learning
Course code	AI-XXX-XXXL
Credit hours	3 (2+1)
Contact hours	2 lecture + 3 lab hours per week
Prerequisites	Machine Learning (or equivalent), Python programming, and basic linear algebra, calculus, probability, and statistics
Environment	Google Colab with PyTorch
	Administrative action Retained from the source outline; the department should replace this placeholder with the approved code.

2. Course description

Deep Learning develops the mathematical and computational foundations of modern neural networks and their responsible application. Students implement, train, regularize, evaluate, explain, and deploy models using PyTorch, covering multilayer networks, convolutional networks, sequence models, transformers, generative models, and graph neural networks. Emphasis is placed on reproducibility, error analysis, computational constraints, fairness, privacy, safety, and environmental impact.

3. Revision priorities

- Sequence concepts from computation and optimization to vision, sequences, transformers, generative modeling, graphs, responsibility, and production.
- Use PyTorch in Google Colab as the common implementation environment for all weekly laboratories.
- Treat the midterm as a separate assessment after Week 8 so that all 16 weeks remain instructionally complete.
- Make evidence visible through predictions, assertions, learning curves, error analysis, saved artifacts, model cards, and reproducible settings.
- Use books only in the formal reference list; web documentation is intentionally excluded from the bibliography.

4. Course learning outcomes

CLO	Learning outcome	Bloom level	PLO mapping
CLO-1	Explain forward and backward computation, optimization behavior, and architectural principles of MLPs, CNNs, sequence models, transformers, generative models, and GNNs.	C2 - Understand	PLO-2
CLO-2	Implement and train deep-learning models in PyTorch for vision, text, time-series, and graph data.	C3 - Apply	PLO-4, PLO-5
CLO-3	Analyze models using appropriate metrics, learning curves, ablation studies, error analysis, and computational cost.	C4 - Analyze	PLO-3, PLO-5

CLO	Learning outcome	Bloom level	PLO mapping
CLO-4	Design a reproducible end-to-end pipeline incorporating transfer learning, tuning, deployment, and monitoring.	C6 - Create	PLO-4, PLO-5
CLO-5	Critically evaluate interpretability, fairness, privacy, safety, environmental, and societal implications and document model limitations.	C5 - Evaluate	PLO-8, PLO-9

5. Teaching and learning approach

- Each 2-hour lecture combines direct instruction, a worked example, an implementation pattern, a diagnostic activity, and an exit ticket.
- Each 3-hour laboratory follows predict-run-measure-explain: students state an expectation, execute a controlled experiment, inspect evidence, and document limitations.
- Datasets are generated inside the notebooks where practical, keeping the baseline credential-free and reducing setup failures.
- A mini-capstone accumulates evidence across comparison, export, latency, drift, explainability, and responsible reporting.

6. Sixteen-week teaching schedule

The reading codes refer to the book-only bibliography in Section 9. Coverage is intentionally cumulative; later weeks reuse evaluation and reproducibility practices introduced early in the course.

Weeks 1-8 | Foundations, vision, and sequence models

Wk	Lecture and key coverage	Three-hour laboratory	CLO / books
1	Deep Learning Workflow and Representation Learning Problem formulation; AI, ML, and DL; tensors and shapes; layered representations; data-model-loss-evaluation loop; reproducibility and compute.	Colab, PyTorch tensors, autograd, and reproducible experiments	CLO-1, CLO-2 B1 Ch. 1-2; B2 Ch. 1-5 selective; B3 Ch. 1-2
2	Multilayer Perceptrons, Activations, and Losses Perceptrons and affine layers; tensor shapes and parameter counts; ReLU, sigmoid, tanh, GELU; regression and classification losses.	MLPs, activations, output-loss pairings, and decision boundaries	CLO-1, CLO-2, CLO-3 B1 Ch. 3-5; B2 Ch. 6; B3 Ch. 4-5
3	Computational Graphs and Backpropagation Reverse-mode automatic differentiation; local derivatives and chain rule; minibatches; initialization; vanishing and exploding gradients; gradient diagnostics.	Manual gradients, autograd verification, and gradient instrumentation	CLO-1, CLO-2, CLO-3 B1 Ch. 6-7; B2 Ch. 6 and 8; B3 Sec. 2.5, 5.3-5.4

Wk	Lecture and key coverage	Three-hour laboratory	CLO / books
4	Optimization, Generalization, and Reliable Training SGD, momentum, RMSProp, Adam; learning-rate schedules; validation design; normalization; dropout; weight decay; augmentation; early stopping; ablation studies.	Optimizer and regularizer ablation with checkpointing	CLO-2, CLO-3, CLO-4 B1 Ch. 8-9 and 20; B2 Ch. 7-8 and 11; B3 Ch. 12 and 19
5	Convolutional Neural Network Foundations Convolution, channels, padding, stride, pooling, output shapes, receptive fields, equivariance, feature maps, and the LeNet-AlexNet-VGG progression.	CNN fundamentals, receptive fields, and feature maps	CLO-1, CLO-2, CLO-3 B1 Ch. 10; B2 Ch. 9; B3 Ch. 7
6	Modern CNNs, Residual Learning, and Transfer Residual, Inception, and DenseNet patterns; batch normalization; augmentation; pretrained backbones; feature extraction; freeze-unfreeze schedules; domain shift.	Residual blocks and synthetic source-to-target transfer learning	CLO-1, CLO-2, CLO-3, CLO-4 B1 Ch. 11; B2 Ch. 9 and 11; B3 Ch. 8 and Sec. 14.1-14.2
7	Object Detection and Semantic Segmentation Classification versus boxes versus masks; IoU, average precision, and Dice; one-stage and two-stage detectors; FCN and U-Net; error analysis.	Synthetic shapes for detection and segmentation	CLO-2, CLO-3, CLO-4 B1 Ch. 10 selective; B2 Ch. 12; B3 Sec. 14.3-14.11
8	Recurrent Networks and Gated Memory Sequence windows; recurrent state; unrolling and backpropagation through time; vanishing and exploding gradients; LSTM, GRU, encoder-decoder patterns, and sequence evaluation.	RNN, LSTM, and GRU time-series forecasting	CLO-1, CLO-2, CLO-3 B2 Ch. 10; B3 Ch. 9-10
		Midterm assessment Schedule the midterm after Week 8 as a separate assessment event. It covers Weeks 1-8 and does not replace a teaching week.	

Weeks 9-16 | Transformers, generation, graphs, responsibility, and deployment

Wk	Lecture and key coverage	Three-hour laboratory	CLO / books
9	Attention Mechanics and Transformer Blocks Queries, keys, and values; scaled dot-product attention; masks; multi-head attention; positional encoding; residual connections; layer normalization; encoder-decoder structure.	Scaled attention and a tiny Transformer	CLO-1, CLO-2, CLO-3 B1 Ch. 12.1-12.4; B3 Ch. 11.1-11.7; B4 Ch. 12
10	Pretraining and Transformer Model Families Tokenization; masked and causal language modeling; BERT, GPT, and T5 patterns; transfer learning; parameter-efficient adaptation concepts; evaluation and limitations.	Toy masked-sequence pretraining and classification transfer	CLO-1, CLO-2, CLO-3, CLO-4 B1 Ch. 12.5-12.10; B3 Sec. 11.9, 15.8-15.10, 16.7
11	Autoencoders and Variational Autoencoders Representation bottlenecks; reconstruction objectives; latent-variable models; ELBO; KL regularization; reparameterization; latent interpolation and posterior collapse.	VAE reconstruction, ELBO, and latent-space exploration	CLO-1, CLO-2, CLO-3 B1 Ch. 14 and 17; B2 Ch. 14 and 20; B4 Ch. 17; B5 Ch. 3

Wk	Lecture and key coverage	Three-hour laboratory	CLO / books
12	Generative Adversarial Networks Minimax objectives; alternating updates; discriminator-generator balance; DCGAN and conditioning; instability; mode collapse; sample evaluation.	Toy GAN training dynamics and mode-collapse diagnosis	CLO-1, CLO-2, CLO-3, CLO-4 B1 Ch. 15; B2 Ch. 20; B4 Ch. 17; B5 Ch. 4
13	Diffusion Models and Latent Generation Forward noising; noise schedules; learned reverse denoising; score/noise prediction; DDPM and DDIM ideas; conditioning; latent diffusion; quality-compute trade-offs.	Toy diffusion with forward and reverse trajectories	CLO-1, CLO-2, CLO-3 B1 Ch. 18; B5 Ch. 8 and 13
14	Graph Neural Networks and Message Passing Graph representation; node, edge, and graph tasks; permutation invariance; message passing; GCN, GraphSAGE, and GAT; inductive learning and oversmoothing.	GCN from first principles on a small explicit graph	CLO-1, CLO-2, CLO-3 B1 Ch. 13; B4 Ch. 13
15	Explainability and Responsible Deep Learning Calibration and shift; saliency, Grad-CAM, LIME, and SHAP; explanation faithfulness; bias sources; subgroup metrics; fairness criteria; privacy, safety, sustainability, and model cards.	Saliency and Grad-CAM with a simulated fairness audit and model card	CLO-3, CLO-5 B1 Ch. 21; B6 Ch. 4-5 and 8; B7 Ch. 6, 8, 11; B8 Ch. 1, 3-5, 9
16	Deployment, Monitoring, and Project Defense Model artifacts and prediction contracts; batch versus online inference; latency, throughput, compression, monitoring, drift, rollback, retraining, reproducibility, and evidence-based project defense.	TorchScript export, latency, drift, and mini-capstone demonstration	CLO-2, CLO-3, CLO-4, CLO-5 B2 Ch. 11; B7 Ch. 6-10

7. Laboratory evidence standard

- An executed notebook with all baseline assertions passing.
- At least one quantitative result and one interpretation tied to the stated prediction.
- Saved figures, metrics, checkpoints, or model artifacts in the designated lab artifact folder.
- A concise limitation or failure-mode statement and a defensible next experiment.
- For responsible-AI tasks, explicit simulation notices and no unsupported real-world fairness claims.

8. Assessment guidance

Policy note Weights below are suggested and should be aligned with current University of Gwadar and departmental assessment regulations before approval.		
Component	Weight	Evidence
Laboratory work and reproducibility checks	20%	16 Colab notebooks, assertions, figures, interpretations, and artifacts
Quizzes and short assignments	10%	Conceptual, mathematical, and diagnostic questions
Midterm examination	20%	Weeks 1-8; scheduled as a separate assessment event
Mini-capstone project	20%	Comparative experiment, model card, artifact, and presentation
Final examination	30%	Cumulative reasoning, model selection, evaluation, and responsible use

9. Reference books

Only books are included. Codes B1-B8 are used throughout the lecture decks, notebooks, and lab manual.

Code	Book	Primary role
B1	Prince, S. J. D. Understanding Deep Learning. 1st ed. The MIT Press, 2023.	Primary modern conceptual text
B2	Goodfellow, I., Bengio, Y., and Courville, A. Deep Learning. 1st ed. The MIT Press, 2016.	Foundational theory and methodology
B3	Zhang, A., Lipton, Z. C., Li, M., and Smola, A. J. Dive into Deep Learning. 1st ed. Cambridge University Press, 2024.	Code, mathematics, and applied examples
B4	Bishop, C. M., and Bishop, H. Deep Learning: Foundations and Concepts. 1st ed. Springer, 2024.	Modern mathematical treatment
B5	Foster, D. Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play. 2nd ed. O'Reilly Media, 2023.	Generative modeling
B6	Thampi, A. Interpretable AI: Building Explainable Machine Learning Systems. 1st ed. Manning, 2022.	Interpretability and explanation
B7	Huyen, C. Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications. 1st ed. O'Reilly Media, 2022.	Deployment, monitoring, and MLOps
B8	Barocas, S., Hardt, M., and Narayanan, A. Fairness and Machine Learning: Limitations and Opportunities. 1st ed. The MIT Press, 2023.	Fairness and responsible machine learning

10. Approval checklist

- ☐ Replace AI-XXX-XXXXL with the approved course code.
- ☐ Confirm CLO-PLO mappings against the current program accreditation matrix.
- ☐ Confirm assessment weights and examination scheduling under current university policy.
- ☐ Confirm Colab/PyTorch access and a fallback CPU environment before semester launch.
- ☐ Review the bibliography annually for new editions while retaining book-only references.